



GATE ME 90 Days Challenge

Day 13/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

**Columns & Energy
Methods**

COUNTDOWN

96 days to GATE

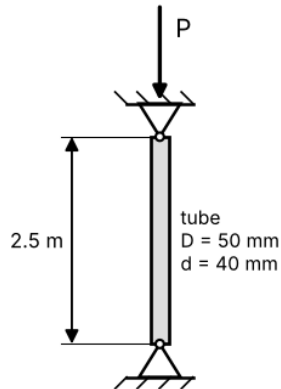
How to use this set: 15 questions (8 MCQ, 7 NAT) on Euler buckling and effective lengths, slenderness ratio, Rankine's formula, strain energy (axial, bending, torsion, impact), Castigliano's theorem, and strain gauges and rosettes. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The steel column shown is a tube of 50 mm outer and 40 mm inner diameter, 2.5 m long, with both ends pinned. For $E = 200$ GPa, its Euler buckling load is nearly



(A) 96.9 kN

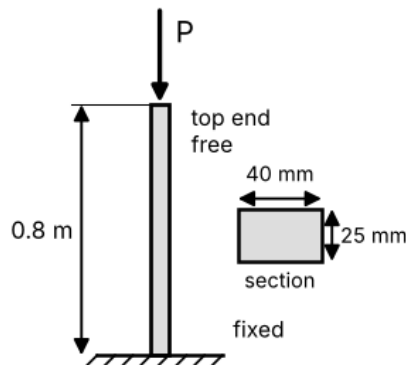
(B) 57.2 kN

(C) 228.8 kN

(D) 14.3 kN

Q2 (NAT)

The aluminium strut shown (40 mm × 25 mm rectangular section, 0.8 m long) is fixed at its base and its top end is not restrained. For $E = 70$ GPa, the Euler buckling load (in kN, round off to two decimal places) is _____

**Q3** (MCQ)

A long steel column with both ends pinned has a slenderness ratio of 120. For $E = 210$ GPa, the Euler critical (crippling) stress is nearly

- (A) 45.8 MPa (B) 575.7 MPa
(C) 143.9 MPa (D) 287.9 MPa

Q4 (NAT)

A solid cast-iron column of 100 mm diameter and 3 m length has both ends pinned. Using Rankine's formula with crushing stress $\sigma_c = 550$ MPa and Rankine constant $a = 1/1600$, the crippling load (in kN) is _____

Q5 (MCQ)

For the same cross-sectional area, material and length, which section gives a column the highest Euler buckling load?

- (A) a solid square (B) a solid circle
(C) a solid rectangle with depth twice the width (D) a thin-walled hollow circular tube

Q6 (NAT)

A solid steel shaft of 50 mm diameter and 1.2 m length transmits a torque of 1.5 kN·m. For $G = 80$ GPa, the strain energy stored in the shaft (in J, round off to two decimal places) is _____

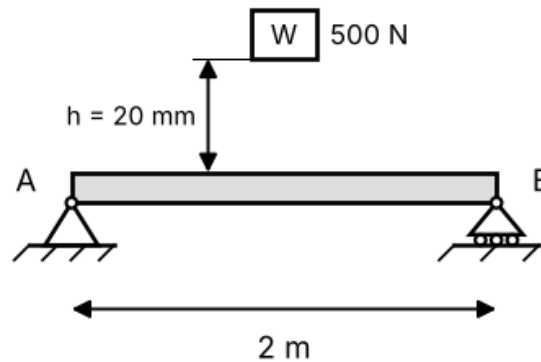
Q7 (MCQ)

An axial load of 30 kN is applied suddenly (no drop height) to a steel bar of cross-sectional area 600 mm² and length 1.5 m. For $E = 200$ GPa, the maximum stress and the maximum elongation are

- (A) 100 MPa and 0.75 mm (B) 50 MPa and 0.375 mm
(C) 100 MPa and 0.375 mm (D) 200 MPa and 1.5 mm

Q8 (NAT)

A 500 N weight is dropped from a height of 20 mm onto the mid-span of the simply supported beam shown (span 2 m, $EI = 200 \text{ kN}\cdot\text{m}^2$). Neglecting the beam's mass and any loss of energy, the maximum deflection at mid-span (in mm, round off to two decimal places) is _____

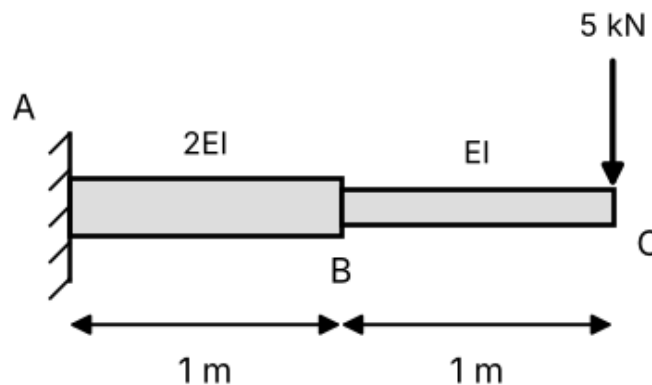
**Q9** (MCQ)

A cantilever of length L and flexural rigidity EI carries a uniformly distributed load w over its whole length. The strain energy stored due to bending is

- (A) $w^2 L^5 / (20EI)$ (B) $w^2 L^5 / (8EI)$
 (C) $w^2 L^5 / (40EI)$ (D) $w^2 L^5 / (48EI)$

Q10 (NAT)

The stepped cantilever ABC shown is fixed at A. Portion AB (1 m) has flexural rigidity $2EI$ and portion BC (1 m) has EI , with $EI = 600 \text{ kN}\cdot\text{m}^2$. Using Castigliano's theorem, the deflection of the end C under the 5 kN load (in mm, round off to two decimal places) is _____

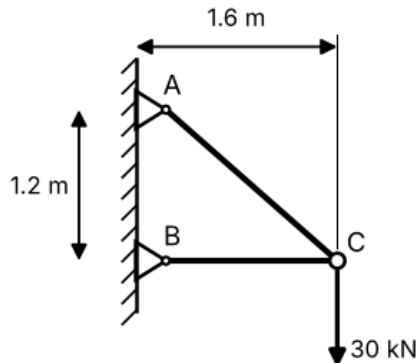
**Q11** (MCQ)

A propped cantilever of span L (fixed at A, propped at B at the same level) carries a UDL w over the whole span. Using the theorem of least work (Castigliano), the reaction at the prop B is

- (A) $5wL/8$ (B) $3wL/8$
 (C) $wL/2$ (D) $5wL/16$

Q12 (NAT)

In the wall bracket shown, the inclined bar AC and the horizontal bar BC are pin-jointed at C and pinned to the wall at A and B (A is 1.2 m above B; BC = 1.6 m). Both bars have cross-sectional area 400 mm^2 and $E = 200 \text{ GPa}$. Using Castigliano's theorem, the vertical deflection of C under the 30 kN load (in mm, round off to two decimal places) is _____

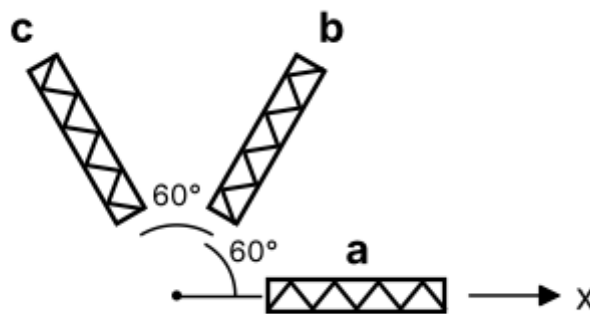
**Q13** (MCQ)

A 45° rectangular rosette reads $\epsilon_0 = 300 \times 10^{-6}$, $\epsilon_{45} = 650 \times 10^{-6}$ and $\epsilon_{90} = 500 \times 10^{-6}$ (angles from the x-axis). The principal strains are

- (A) 650×10^{-6} and 300×10^{-6} (B) 1057.6×10^{-6} and -257.6×10^{-6}
 (C) 909.9×10^{-6} and -109.9×10^{-6} (D) 669.3×10^{-6} and 130.7×10^{-6}

Q14 (NAT)

A 60° delta rosette (figure) on an aluminium plate reads $\epsilon_a = 800 \times 10^{-6}$, $\epsilon_b = 200 \times 10^{-6}$ and $\epsilon_c = 500 \times 10^{-6}$. For $E = 70 \text{ GPa}$ and $\nu = 0.33$, the major principal stress (in MPa, round off to two decimal places) is _____

**Q15** (MCQ)

A single strain gauge is bonded on the surface of a shaft in pure torsion, at 45° to the shaft axis. It reads 400×10^{-6} . For $G = 80 \text{ GPa}$, the maximum shear stress at the surface is

- (A) 64 MPa (B) 32 MPa
 (C) 128 MPa (D) 80 MPa

Answer Key

Q	Type	Answer
Q1	MCQ	(B) 57.2 kN
Q2	NAT	14.06 (accept 13.95 to 14.15)
Q3	MCQ	(C) 143.9 MPa
Q4	NAT	432.0 (accept 429 to 435)
Q5	MCQ	(D) a thin-walled hollow circular tube
Q6	NAT	27.50 (accept 27.3 to 27.7)
Q7	MCQ	(A) 100 MPa and 0.75 mm
Q8	NAT	4.52 (accept 4.48 to 4.56)
Q9	MCQ	(C) $w^2L^5/(40EI)$
Q10	NAT	12.50 (accept 12.4 to 12.6)
Q11	MCQ	(B) $3wL/8$
Q12	NAT	3.15 (accept 3.12 to 3.18)
Q13	MCQ	(D) 669.3×10^{-6} and 130.7×10^{-6}
Q14	NAT	70.47 (accept 70.0 to 70.9)
Q15	MCQ	(A) 64 MPa

Detailed Solutions

Q1 Answer: (B) 57.2 kN

Concept: Euler load $P_{cr} = \pi^2 EI / L_e^2$, with $L_e = L$ for both ends pinned.

$$I = \pi(D^4 - d^4)/64 = \pi(50^4 - 40^4)/64 = \mathbf{1.811 \times 10^5 \text{ mm}^4}$$

$$P_{cr} = \pi^2 \times 200000 \times 1.811 \times 10^5 / 2500^2 = 57\,207 \text{ N} = \mathbf{57.2 \text{ kN}}$$

Exam tip: 228.8 kN treats the ends as fixed ($L_e = L/2$) and 14.3 kN as fixed at one end with the other end free ($L_e = 2L$). Read the end conditions first.

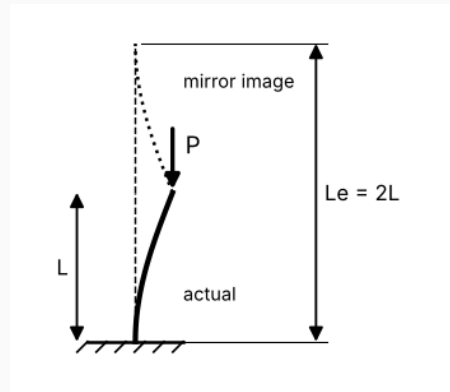
Q2 Answer: 14.06 (accept 13.95 to 14.15)

Concept: One end fixed, other end free $\rightarrow L_e = 2L$. Buckling occurs about the axis with the least I .

$$L_e = 2 \times 0.8 = \mathbf{1.6 \text{ m}}; I_{\min} = 40 \times 25^3/12 = \mathbf{52\,083 \text{ mm}^4}$$

$$P_{cr} = \pi^2 \times 70000 \times 52\,083 / 1600^2 = 14\,056 \text{ N} = \mathbf{14.06 \text{ kN}}$$

Exam tip: The deflected shape of a column fixed at the base and free at the top is one quarter of a sine wave; its mirror image completes the half wave of a pinned column of length $2L$ (figure).

**Q3 Answer: (C) 143.9 MPa**

Concept: $\sigma_{cr} = P_{cr}/A = \pi^2 EI / (AL_e^2) = \pi^2 E / (L_e/k)^2$, since $I = Ak^2$.

$$\sigma_{cr} = \pi^2 \times 210000 / 120^2 = 2\,072\,617 / 14\,400 = \mathbf{143.9 \text{ MPa}}$$

This is below the usual proportional limit of structural steel (about 200 to 250 MPa), so Euler's formula is valid here.

Exam tip: 45.8 MPa uses π instead of π^2 . Critical stress depends only on E and slenderness, not on the absolute size.

Q4 Answer: 432.0 (accept 429 to 435)

Concept: Rankine: $P = \sigma_c A / [1 + a(L_e/k)^2]$. It blends crushing (short columns) and buckling (long columns).

$$A = \pi \times 100^2/4 = \mathbf{7854 \text{ mm}^2}; k = d/4 = \mathbf{25 \text{ mm}}; L_e/k = 3000/25 = \mathbf{120}$$

$$1 + a(L_e/k)^2 = 1 + 14\,400/1600 = \mathbf{10}$$

$$P = 550 \times 7854 / 10 = 431\,969 \text{ N} = \mathbf{432.0 \text{ kN}}$$

Exam tip: For a solid circle $k = d/4$; for a rectangle $k_{\min} = b/\sqrt{12}$. The crushing load alone (4320 kN) is ten times higher.

Q5 Answer: (D) a thin-walled hollow circular tube

Concept: $P_{cr} = \pi^2 E A k_{\min}^2 / L_e^2$. With A fixed, the section with the largest least radius of gyration wins.

Pushing material away from the centroid raises $I = A k^2$ without adding area. A tube does this equally about every axis, so it has no weak axis.

Rectangles buckle about their weak axis; a solid circle or square keeps material near the centroid.

Answer: (D) the hollow circular tube.

Exam tip: This is why scaffold poles and many struts are tubes. Very thin walls can, however, fail by local buckling.

Q6 Answer: 27.50 (accept 27.3 to 27.7)

Concept: Torsional strain energy $U = T^2L/(2GJ)$ (area under the linear T - θ graph).

$$J = \pi \times 50^4/32 = \mathbf{6.136 \times 10^5 \text{ mm}^4}$$

$$U = (1.5 \times 10^6)^2 \times 1200 / (2 \times 80000 \times 6.136 \times 10^5) = 27\,502 \text{ N}\cdot\text{mm} = \mathbf{27.50 \text{ J}}$$

Exam tip: Equivalent form: $U = (\tau_{\max}^2/4G) \times \text{volume}$ for a solid shaft. Keep units in N and mm, then divide by 1000 for J.

Q7 Answer: (A) 100 MPa and 0.75 mm

Concept: For a suddenly applied load, work done $P \cdot \delta$ equals strain energy $\sigma^2AL/(2E)$, which gives $\sigma = 2P/A$ (twice the static value).

$$\sigma = 2 \times 30000/600 = \mathbf{100 \text{ MPa}}$$

$$\delta = \sigma L/E = 100 \times 1500/200000 = \mathbf{0.75 \text{ mm}}$$

Exam tip: 50 MPa and 0.375 mm are the gradual-load values. A sudden load doubles both stress and elongation.

Q8 Answer: 4.52 (accept 4.48 to 4.56)

Concept: Energy balance $W(h + \delta) = \frac{1}{2}k\delta^2$, with beam stiffness $k = W/\delta_{\text{st}}$. This gives $\delta = \delta_{\text{st}}[1 + \sqrt{1 + 2h/\delta_{\text{st}}}]$.

$$\delta_{\text{st}} = WL^3/(48EI) = 500 \times 2000^3 / (48 \times 200 \times 10^9) = \mathbf{0.4167 \text{ mm}}$$

$$\text{Impact factor} = 1 + \sqrt{1 + 2 \times 20/0.4167} = 1 + \sqrt{97} = \mathbf{10.85}$$

$$\delta_{\text{max}} = 10.85 \times 0.4167 = \mathbf{4.52 \text{ mm}}$$
 (equivalent static load about 5.42 kN)

Exam tip: The same impact-factor formula works for bars, beams and springs. Only the static deflection changes.

Q9 Answer: (C) $w^2L^5/(40EI)$

Concept: $U = \int M^2/(2EI) \, dx$ over the length.

From the loaded tip, $M = wx^2/2$, so $M^2 = w^2x^4/4$

$$U = \int_0^L w^2x^4/(8EI) \, dx = w^2L^5/(40EI)$$

Answer: (C) $w^2L^5/(40EI)$.

Exam tip: Compare an end point load P : $M = Px$ gives $U = P^2L^3/(6EI)$. Integrate M^2 over the length; do not just multiply the end moment by the length.

Q10 Answer: 12.50 (accept 12.4 to 12.6)

Concept: $\delta_C = \partial U/\partial P = \int M(\partial M/\partial P) \, dx/EI$, taken segment by segment with each segment's own rigidity.

Measure x from C: $M = Px$, $\partial M/\partial P = x$.

$$\text{BC (0 to 1 m, EI): } \int Px^2 \, dx/EI = P/(3EI)$$

$$\text{AB (1 to 2 m, 2EI): } \int Px^2 \, dx/(2EI) = P(8 - 1)/(6EI) = 7P/(6EI)$$

$$\delta_C = (1/3 + 7/6)P/EI = 1.5 \times 5/600 = 0.0125 \text{ m} = \mathbf{12.50 \text{ mm}}$$

Exam tip: A uniform cantilever with EI throughout would give $P(2)^3/(3EI) = 22.2 \text{ mm}$. Stiffening the root section matters most because M is largest there.

Q11 Answer: (B) $3wL/8$

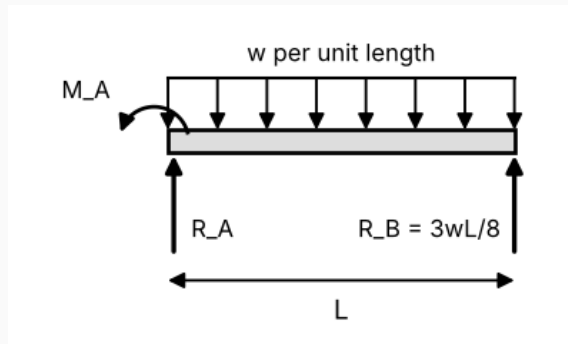
Concept: Treat the prop reaction R as redundant. Since the prop does not move, $\partial U/\partial R = 0$.

From B: $M = Rx - wx^2/2$, $\partial M/\partial R = x$

$$\int_0^L (Rx - wx^2/2)x \, dx = 0 \rightarrow RL^3/3 - wL^4/8 = 0$$

$R = 3wL/8$; then $R_A = 5wL/8$ and $M_A = wL^2/8$ (FBD below).

Exam tip: $5wL/8$ is the fixed-end reaction, a common swap. $5P/16$ is the prop reaction for a central point load.

**Q12 Answer: 3.15 (accept 3.12 to 3.18)**

Concept: For a pin-jointed frame, $\delta = \Sigma F(\partial F/\partial P)L/(AE) = \Sigma F^2L/(PAE)$ when every F is proportional to P .

$AC = \sqrt{(1.6^2 + 1.2^2)} = 2.0 \text{ m}$. Joint C: $F_{AC} \times 1.2/2.0 = 30 \rightarrow F_{AC} = 50 \text{ kN (tension)}$; $F_{BC} = 50 \times 1.6/2.0 = 40 \text{ kN (compression)}$

$$\delta = (50000^2 \times 2000 + 40000^2 \times 1600) / (30000 \times 400 \times 200000)$$

$$\delta = (5.0 \times 10^{12} + 2.56 \times 10^{12}) / (2.4 \times 10^{12}) = 3.15 \text{ mm}$$

Exam tip: Signs do not matter in F^2L terms, but they do when you use a unit load in a different direction from the real load.

Q13 Answer: (D) 669.3×10^{-6} and 130.7×10^{-6}

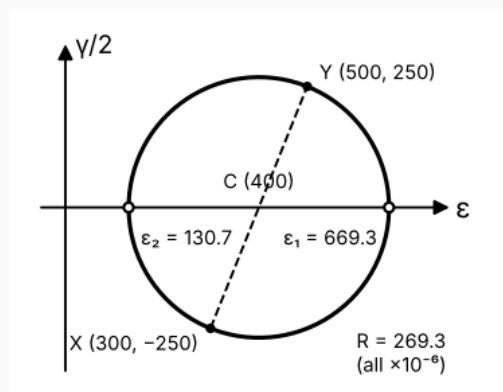
Concept: For a 0° - 45° - 90° rosette: $\epsilon_x = \epsilon_0$, $\epsilon_y = \epsilon_{90}$, $\gamma_{xy} = 2\epsilon_{45} - \epsilon_0 - \epsilon_{90}$.

$$\gamma_{xy} = 1300 - 300 - 500 = 500 \times 10^{-6}$$

Centre = $(300 + 500)/2 = 400$; radius = $\sqrt{[(300 - 500)/2]^2 + (500/2)^2} = \sqrt{(100^2 + 250^2)} = 269.3$

$\epsilon_{1,2} = 400 \pm 269.3 = 669.3 \times 10^{-6}$ and 130.7×10^{-6} ; $\tan 2\theta_p = 500/(-200)$ gives ϵ_1 at 55.9° from the x-axis.

Exam tip: Gauges measure normal strain only. Shear strain always comes from the combination of the three readings, never from the 45° gauge alone.



Q14 Answer: 70.47 (accept 70.0 to 70.9)

Concept: For a 0° - 60° - 120° rosette: centre = $(\epsilon_a + \epsilon_b + \epsilon_c)/3$ and radius = $\sqrt{[(\epsilon_a - \text{centre})^2 + ((\epsilon_b - \epsilon_c)/\sqrt{3})^2]}$.

Centre = $(800 + 200 + 500)/3 = 500$; radius = $\sqrt{(300^2 + (-300/\sqrt{3})^2)} = \sqrt{(90\,000 + 30\,000)} = 346.4$

$\epsilon_1 = 846.4 \times 10^{-6}$, $\epsilon_2 = 153.6 \times 10^{-6}$

$\sigma_1 = E(\epsilon_1 + \nu\epsilon_2)/(1 - \nu^2) = 70000 \times (846.4 + 0.33 \times 153.6) \times 10^{-6} / 0.8911 = 70.47 \text{ MPa}$

Exam tip: $\sigma_1 \neq E\epsilon_1$ (that gives 59.2 MPa). The Poisson coupling with ϵ_2 must be included.

Q15 Answer: (A) 64 MPa

Concept: Pure shear: the principal directions are at 45° to the axis, and $\epsilon_{45} = \gamma/2$ (the principal strain).

$\gamma = 2 \times 400 \times 10^{-6} = 800 \times 10^{-6}$

$\tau = G\gamma = 80000 \times 800 \times 10^{-6} = 64 \text{ MPa}$

Exam tip: 32 MPa forgets that ϵ_{45} is only half of γ . A gauge along the shaft axis would read zero in pure torsion.

What next?

Day 13 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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Next, Day 14/90: Theory of Machines, Mechanism Analysis. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.